D1 : Simulation of system with ideal hardware

Wireless Communication Systems

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1 D1.1 Ideal Simulation

2 D1.2 Specific deliverable objectives

Coding part

- We consider a BCH code of length $N_c = 31$. We have $GF(32) = \frac{GF(2)[x]}{1+x^2+x^5}$. Let $\varphi(x) = 1 + x^2 + x^5$ be the characteristic polynomial of $GF(32)$.

We will first consider that the code corrects $t = 1$ error.

2.1 Case without coding and with a code correcting $t = 1$ error

Let k be the number of information bits such that the code rate is given by $R = \frac{k}{n}$.

We have $g(x) = LCM(\mu_\alpha, \mu_{\alpha^2})$. Since α is a root of $\varphi(x)$ (primitive element of $GF[32]$) and the code having $2t = 2$ consecutive zeros, α^2 is also a root of $\varphi(x)$.

Thus $\mu_\alpha = \mu_{\alpha^2} = 1 + x^2 + x^5 = g(x)$. We therefore observe that $\deg(g(x)) = n - k = 5$ with $n = 31$, giving $k = 26$.

Therefore <u>R =26/31</u>

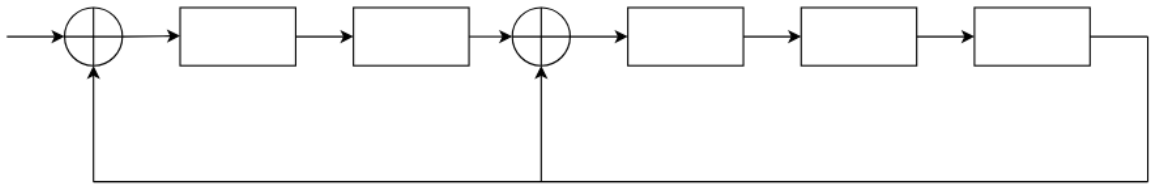

Figure 1: Systematic BCH encoder
$g(x) = 1 + x^2 + x^5$

Since the codeword is expressed as:

$$c(x) = x^{n-k} \times m(x) + x^{n-k} \times m(x) \bmod (g(x))$$

To compute it, it must be split into two parts. One that simply shifts the bits of the information word $m(x)$ by $n - k$ bits to the left (in our representation, MSBs are on the left and LSBs on the right).

The modulo computation must then be built, which can be represented by a systematic encoder with 5 shift registers as shown in Figure 1.

We implement the functions “calcul_crc.m” and “verif_modulo.m”. Comparing the two results allows verification of the correct modulo computation. “calculer_crc_fast.m” is an optimized version for subsequent computations.

Decoding of the codeword is theoretically performed using maximum likelihood decoding.

In a Binary Symmetric Channel (BSC), each bit has a probability p of being flipped. The probability of receiving $m(x)$ given that c was transmitted depends only on the number of erroneous bits, i.e., the Hamming weight of the error $w(e)$. The codeword estimated by the decoder is:

$$\hat{c} = \arg\min d_{\mathrm{H}}\ (m, c)$$

Maximum likelihood decoding, while optimal, quickly becomes complex to implement for long codes as it requires comparing the received word to all possible codewords. For a BCH(31,26) code, this represents 2^{26} comparisons. Syndrome decoding is an efficient implementation of this principle; it exploits the algebraic structure of the code to reduce the search to a simple table lookup, while guaranteeing the same optimal result in the case of correctable errors.

The BCH code with t=1 is capable of correcting a single error. It is defined by $g(x)$ the generator polynomial such that the syndrome at reception is:

$$S(x) = m(x) \bmod g(x)$$

Since $C(x) \equiv 0 \bmod g(x)$, then:

$$S(x) = e(x) \bmod g(x)$$

The syndrome therefore depends only on the error. In the case t = 1, the error is of the form:

$$e(x) = x^i$$

There is therefore a correspondence such that each error position produces a different syndrome. Syndrome decoding is therefore performed.

The code "Build_syndrome_table_t1.m" builds a reference table of all possible syndromes for the t=1 case. It computes the syndrome of each single error (positions 1 to 31) by passing each error vector e through the shift register "Registre_decalage_modulo.m".

Then "Décodeur_t1.m" computes the syndrome of the received word r(x), encoded by "calculer_crc.m", and searches for this syndrome in the table. If a match is found, the error position is identified and the bit is corrected by a simple XOR.

If the syndrome is zero, no error is detected and the received word r(x) is considered valid.

In order to compare the performance between the different codes, we aim to plot their Bit Error Rate (BER) as a function of the normalized SNR Eb/N_0.

The propagation channel is modeled by additive white Gaussian noise.

We know that $\frac{Eb}{N0} = \frac{Es}{R \log 2(C) N_0}$ with:

- Es the symbol energy, taken equal to 1 Joule for example
- R the code rate
- C the constellation size
- N_0 the variance of the Gaussian channel noise

First, we consider an uncoded BPSK constellation with R = 1 and C = 2. We have:

$$\left(\frac{Eb}{N_0}\right)_{\text{dB}} = 10\log\left[\frac{Es}{R\log 2(C)}\right] - 10\log[N_0]$$

that is

$$N_0 = 10^{\log\left[\frac{Es}{R\log 2(C)} - \frac{1}{10}\left(\frac{Eb}{N_0}\right)_{\text{dB}}\right]}$$

For a given $\frac{Eb}{N_0}$, we model a channel and therefore a different BER. The theoretical performance of the code is furthermore given by:

$$BER = Q\left(\sqrt{\frac{2EbR}{N_0}}\right)$$

Using the code "BPSK_performance_plotting.m", we obtain the following curve:

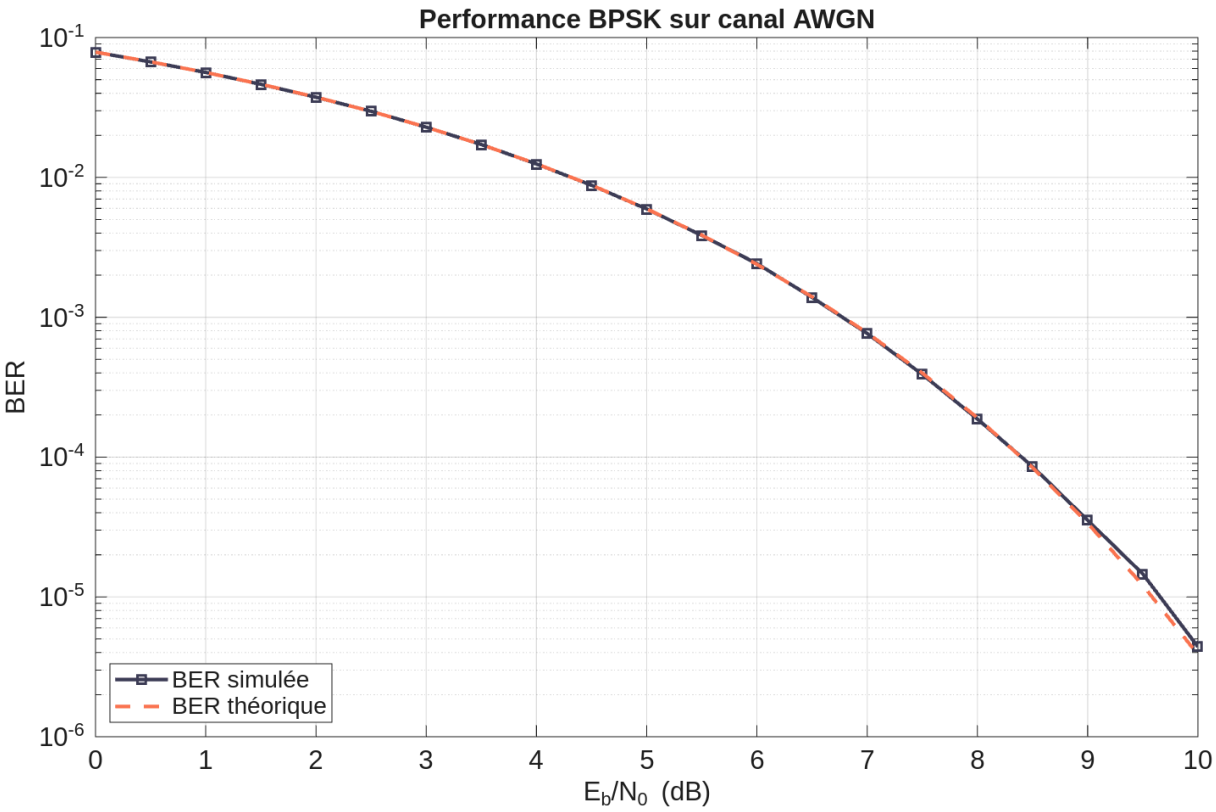

Figure 2: BPSK performance on AWGN channel

Without coding, the practical simulation follows the theoretical prediction.

Let us now add coding and decoding by the BCH code above. For t = 1 error, we now have R = 26/31. We encode again with "calcul_crc_fast.m" and use "Decodeur_t1_batch.m" (optimized version of "Decodeur_t1.m") to recover $\hat{c}$.

After BPSK demodulation on the AWGN channel, the BER on each received coded bit is:

$$p = Q\left(\sqrt{\frac{2REb}{N_0}}\right)$$

A codeword contains j errors such that: $P_{\text{j errors}} = \binom{n}{j} p^{\text{j}}\ (1-p)^{\text{n-j}}$

The BCH decoder (n,k,t) corrects for j $\leq$ t errors, so its failure probability is given by

$$P_{\text{fail}} = \sum_{j=t+1}^{n} \binom{n}{j} p^{\text{j}}\ (1-p)^{\text{n-j}} = 1 - \sum_{j=0}^{t} \binom{n}{j} p^{\text{j}}\ (1-p)^{\text{n-j}}$$

Finally, considering errors uniformly distributed over the k information bits, we obtain

$$BER_{\text{BCH}} = \frac{1}{k} \sum_{j=t+1}^{n} \binom{n}{j} p^{\text{j}}\ (1-p)^{\text{n-j}}$$

To limit simulation times, the program "BCH1_performance_plotting_fast.m" sets thresholds on the minimum number of errors as well as the minimum number of words to compute. This induces a slight deviation from reality but greatly shortens the simulation time.

We then obtain the following curve:

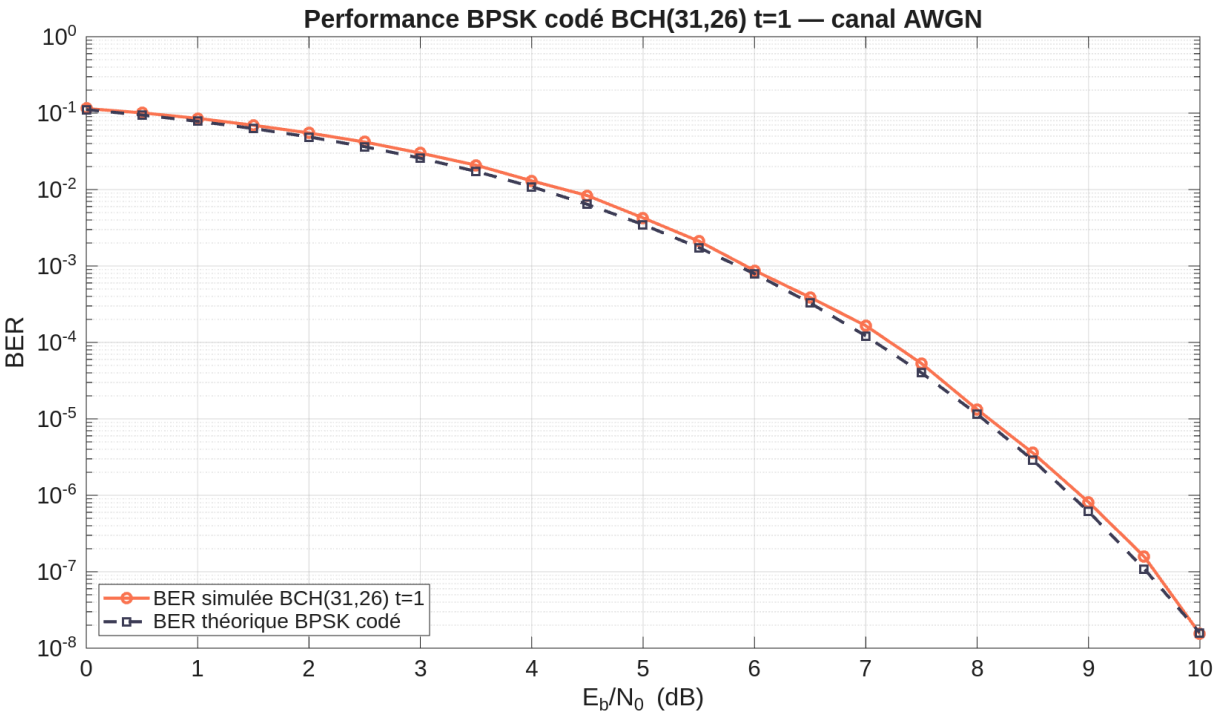

Figure 3: BPSK coded BCH(31,26) performance correcting t = 1 - AWGN channel

Again, the simulation with coding remains faithful to the theoretical computations. At high SNR, the number of points, and thus the approximations, become slightly more significant.

2.2 Case $t = 2$ errors

Let us now consider that the code corrects $t = 2$ errors.

The roots of φ are $\alpha, \alpha^2, \alpha^3$ and α^4. We compute the minimal polynomials:

- $\mu_\alpha = (x-\alpha)(x-\alpha^2)(x-\alpha^4)(x-\alpha^8)(x-\alpha^{16}) = 1 + x^2 + x^5$
- We note that α^2 and α^4 are already in the conjugacy class of α, so we do not need to compute their minimal polynomial.
- $\mu_{\alpha^3} = (x-\alpha^3)(x-\alpha^6)(x-\alpha^{12})(x-\alpha^{24})(x-\alpha^{17}) = 1 + x^2 + x^3 + x^4 + x^5$

Then, $g(x) = LCM(\mu_\alpha, \mu_{\alpha^3}) = \mu_\alpha\ \mu_{\alpha^3}$, the minimal polynomials being coprime.

Finally, $g(x) = g(x) = 1 + x^3 + x^5 + x^6 + x^8 + x^9 + x^{10}$

$\deg(g(x)) = 10 = n - k$ giving $k = 21$ and thus <u>R =21/31</u>.

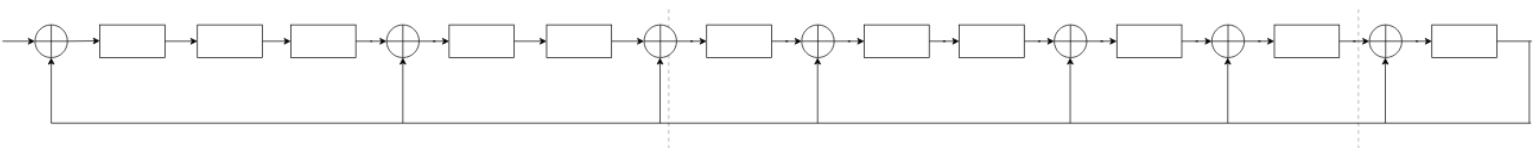

Figure 4: Systematic BCH encoder
$g(x) = 1 + x^3 + x^5 + x^6 + x^8 + x^9 + x^{10}$

The decoding principle for the t=2 case is the same as for t=1, except that the error is theoretically expressed as:

$$e(x) = x^{i_1} + x^{i_2}$$

And the syndromes as:

$$S_1 = r(x) \bmod \mu_\alpha = \alpha^{i_1} + \alpha^{i_2}$$

$$S_3 = r(x) \bmod \mu_{\alpha^3} = \alpha^{3i_1} + \alpha^{3i_2}$$

However in "Build_syndrome_table_t2.m", we do not compute the two syndromes, as this requires solving a system in GF(2^5). The code creates an exhaustive table of all possible errors for t=2, such that:

$$S = e(x) \bmod G_2(X)$$

The syndrome depends only on the error, not on the transmitted word. Although the classical algebraic approach requires two syndromes S1 and S3 to solve a system with two unknowns, the exhaustive table approach allows the use of a single syndrome S.

This is possible because G2(x) is constructed such that all 496 error patterns produce distinct syndromes, guaranteeing a unique correspondence between a received syndrome and an error pattern.

All possible syndromes are precomputed for all error patterns, yielding a table of 31 single errors and 465 double errors. The resulting table has dimensions of 496 × 12. In "Build_syndrome_table_t2.m" the table is organized such that:

- Column 1: position i_1 of the first error
- Column 2: position i_2 of the second error (0 if single error)
- Columns 3 to 12: syndrome S on 10 bits

Upon reception, "Décodeur_t2.m" computes S, searches for this syndrome in the table, and corrects the bits at the found positions by XOR, exactly as for t=1.

By changing the generator polynomial and the code rate in the previously used MATLAB codes, we obtain the following performances:

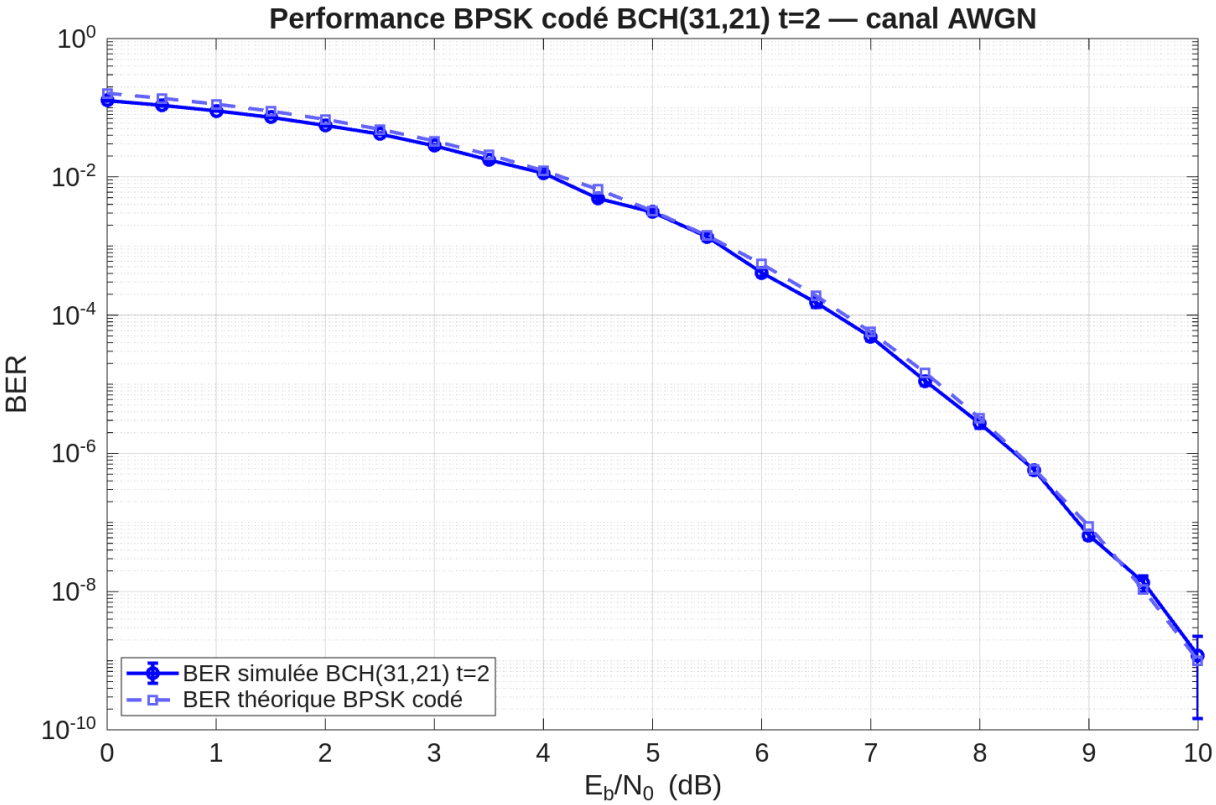

Figure 5: BPSK coded BCH(31,21) performance correcting t = 2 errors - AWGN channel

The computations being significant, a slight divergence is observed at very high SNR. The addition of uncertainty bars related to the simulation nevertheless confirms its reliability.

Finally, we can consolidate the performance with and without coding:

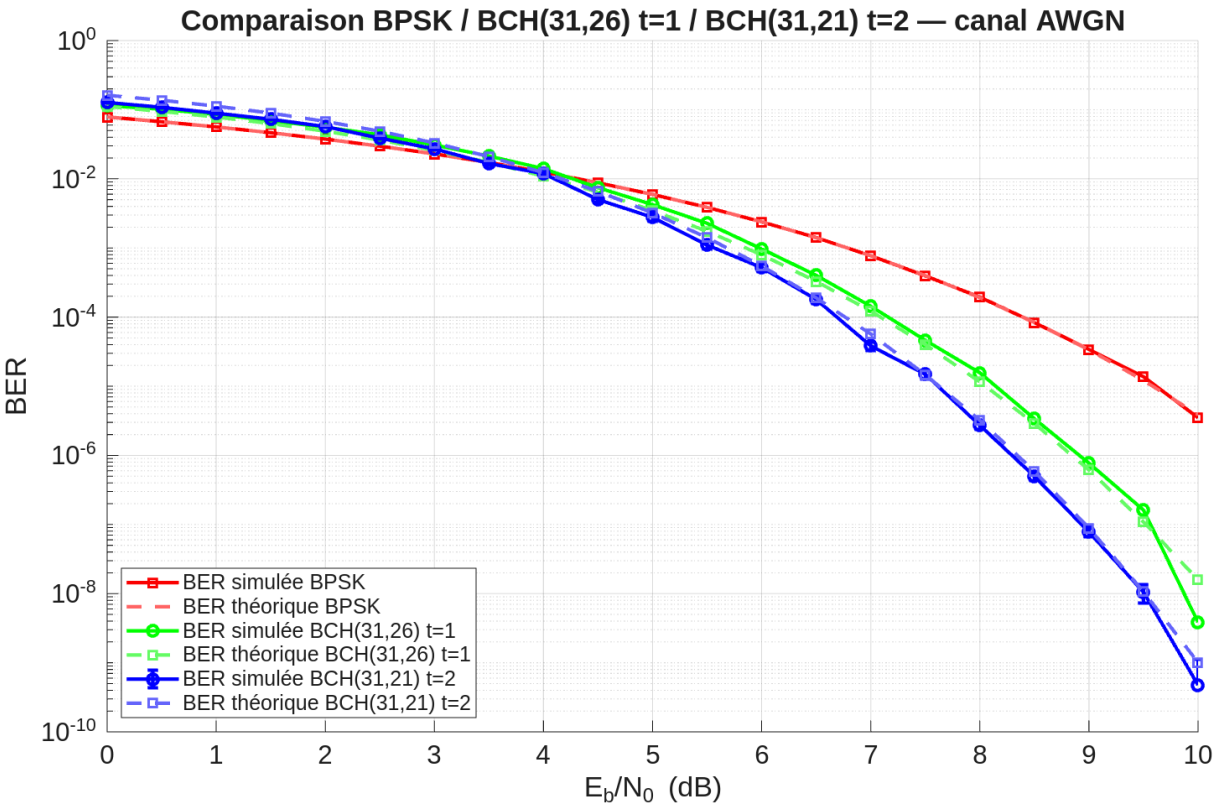

Figure 6: Performance comparison by coding scheme

At low SNR, the coding and decoding steps introduce more approximation than they improve BER. However, as expected, error correction through coding reduces BER at high SNR.

3 Reception Performance Analysis: Equalization and Channels

In this section, we study the impact of different propagation channels on a BPSK-modulated signal, as well as the ability of several equalizers to correct Inter-Symbol Interference (ISI).

3.1 Analytical Modeling and Channel Performance Predictions

The transmission is modeled in matrix form by the following equation:

$$z = Hs + w$$

where z is the vector of received symbols, s the vector of transmitted symbols, w the additive white Gaussian noise with variance N_0, and H the Toeplitz matrix modeling the impulse response of the channel.

We evaluate the system on 3 specific channels:

- **Channel 1 (Weakly selective)**: $A = [1, 0.1, 0.1, 0.1, 0.1]$. The echoes being very weak, the matrix H is diagonally dominant. ISI is minimal. We can therefore assume that linear equalizers will suffice to achieve excellent performance.

- **Channel 2 (Moderately selective)**: $A = [1, 0.8, 0.6, 0.4, 0.2]$. The sum of echo amplitudes exceeds that of the main path. A simple threshold detector is likely to fail, and channel inversion will begin to amplify noise.
- **Channel 3 (Highly selective)**: $A = [1, 0.8, 0.8, 0.8, 0.8]$. This channel exhibits deep fading. The matrix H will be very poorly conditioned. We can predict a collapse of linear equalizers due to strong noise amplification.

3.2 Channel Modeling and Visualization

Before evaluating reception performance, it is essential to analyze the discrete impulse response of the different simulated channels. The overall continuous channel $\tilde{h}(t)$ models the cascade of the shaping filter (raised cosine with **roll-off** $\rho = 0.5$) and the multipath propagation channel. The discrete impulse response $h(m)$ is obtained by sampling at the symbol rate $T_s = 0.05\mu s$:

$$h(m) = \tilde{h}(mT_s) = \sum_{k=0}^{M} A_k \tilde{g}(mT_s - \tau)$$

The resulting filter is then normalized so that its total energy is unity ($\sum \|h(m)\|^2 = 1$). The figures below illustrate the shape of $h(m)$ for the three multipath configurations studied.

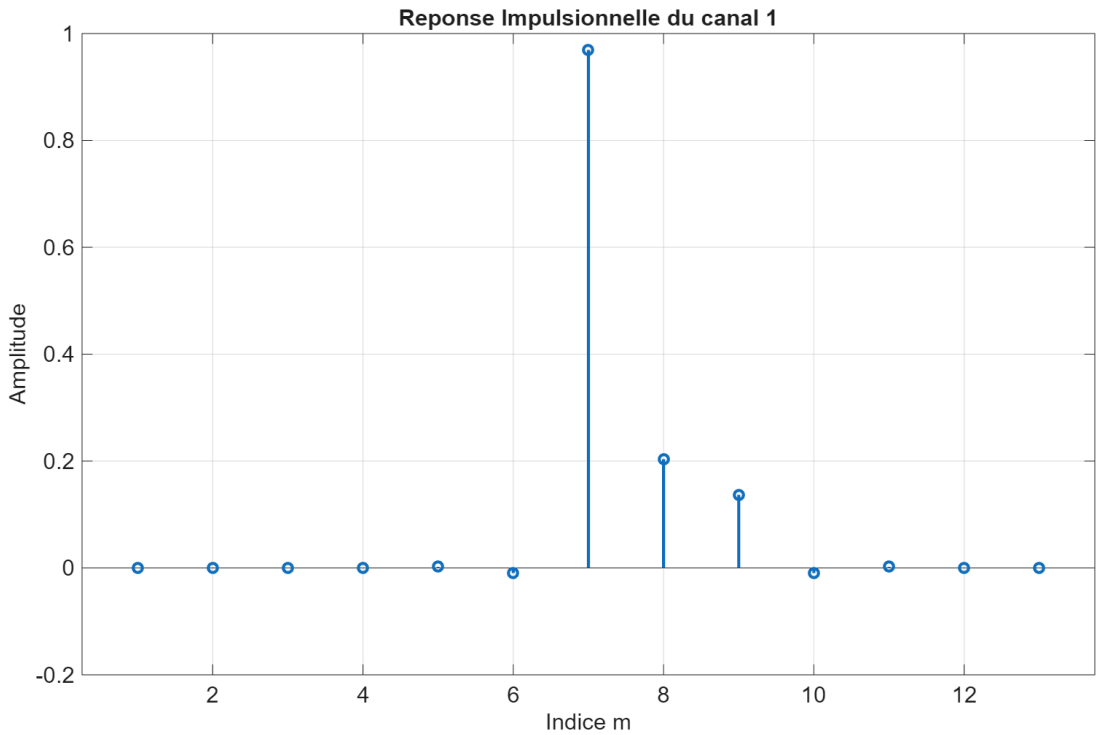

Figure 7: Discrete impulse response $h(m)$ for Channel 1

Interpretation (Channel 1): This channel features a dominant main path ($A_0 = 1$) followed by four very weak echoes ($A_k = 0.1$). Sampling the continuous response produces a central coefficient very close to 1, with the other coefficients (representing Inter-Symbol Interference - ISI) remaining minimal. This is an "easy" Line-of-Sight channel that barely disturbs the signal.

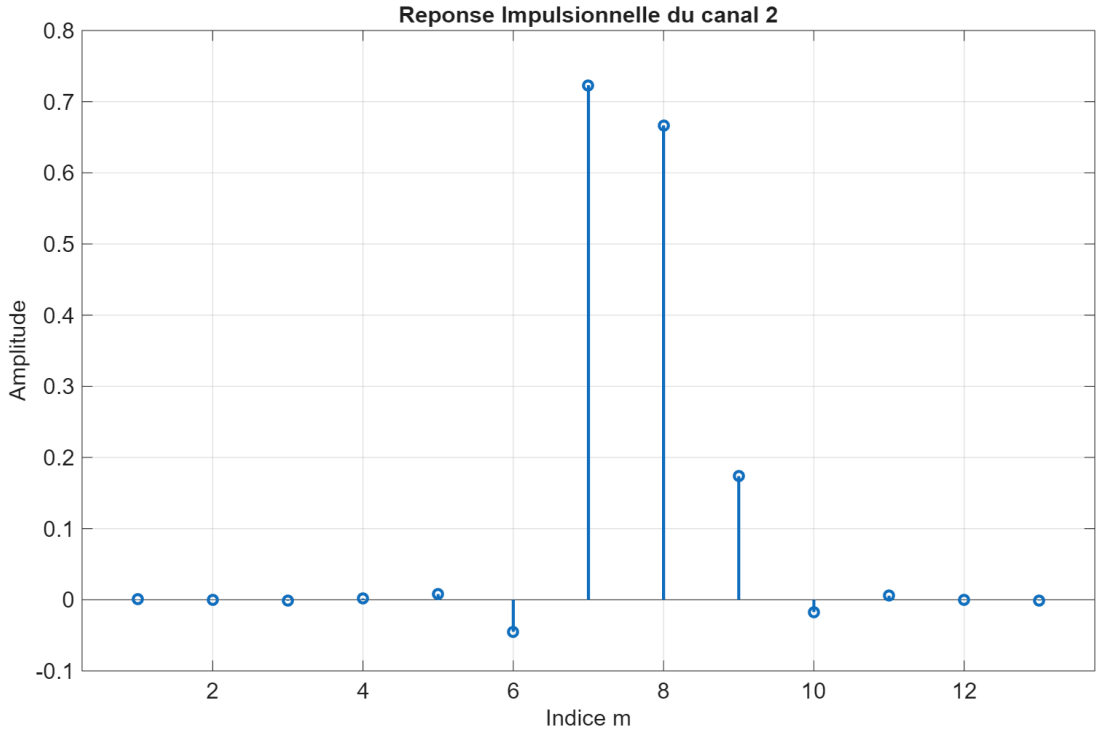

Figure 8: Discrete impulse response $h(m)$ for Channel 2

Interpretation (Channel 2): For this channel, the echoes have linearly decreasing attenuation ($A = [1, 0.8, 0.6, 0.4, 0.2]$). Since the delays τ_k are fractional multiples of the symbol time ($0.5T_s, 1.5T_s...$), sampling the Nyquist pulse combines the energy from different paths across several adjacent coefficients. A clear temporal spreading of energy is observed: ISI becomes significant, requiring specific processing at reception.

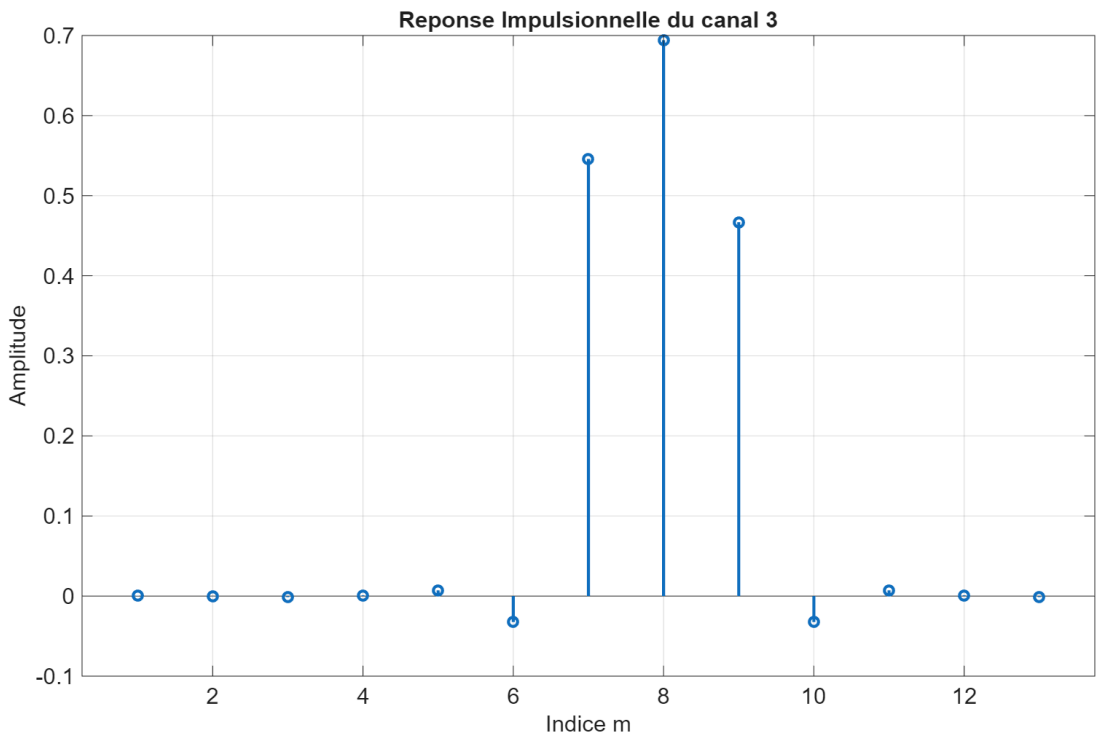

Figure 9: Discrete impulse response $h(m)$ for Channel 3

Interpretation (Channel 3): This channel represents a very hostile environment where all echoes have strong and constant amplitude ($A_k = 0.8$). The figure shows that the signal energy is heavily dispersed over several symbols, with secondary coefficients whose amplitude rivals that of the main coefficient. In the frequency domain, this results in deep fading (spectral notches), making the channel matrix particularly difficult to invert.

3.3 Mathematical Models of Equalizers

To counter ISI, we implement three receivers:

1. The Threshold Detector

This detector ignores the effect of the channel and makes a direct hard decision on the received signal. For BPSK modulation, the decision is:

$$\hat{s}_{\text{threshold}} = \text{sign}(\text{Re}(z))$$

For multipath channels (Channels 2 and 3), this equalizer is destined to fail as it does not handle ISI.

2. The Zero-Forcing (ZF) Equalizer

The ZF aims to completely invert the channel by applying H^{-1} to the received signal, the Toeplitz matrix H being square $N \times N$:

$$z_{\text{ZF}} = H^{-1}z = s + H^{-1}w$$

$$\hat{s}_{\text{ZF}} = \text{sign}(\text{Re}(z_{\text{ZF}}))$$

While this method drastically cancels ISI, its major drawback lies in the term $H^{-1}w$. If H has small eigenvalues (as for Channel 3), H^{-1} will have very large values, significantly amplifying the noise.

3. The Decision Feedback Equalizer (DFE)

The DFE uses the QR decomposition of the channel matrix: $H = QR$, with Q unitary and R upper triangular. The reception equation becomes:

$$z = QRs + w \Longrightarrow \tilde{z} = Q^H z = Rs + Q^H w$$

Since R is upper triangular, the symbol $s(k)$ only suffers interference from "future" symbols $s(k+1)...s(N)$. The DFE decodes the block in reverse (from N to 1) by subtracting the interference computed from the **already decided** symbols $\hat{s}$.
The soft estimate before decision is written as:

$$s_{\text{soft}(k)} = \frac{\tilde{z}(k) - \sum_{j=k+1}^{N} R(k,j)\hat{s}(j)}{R(k,k)}$$

By avoiding the direct inversion of H, the DFE bypasses noise amplification and should offer the best performance on difficult channels.

3.4 Simulation Results and Interpretations (BPSK Modulation)

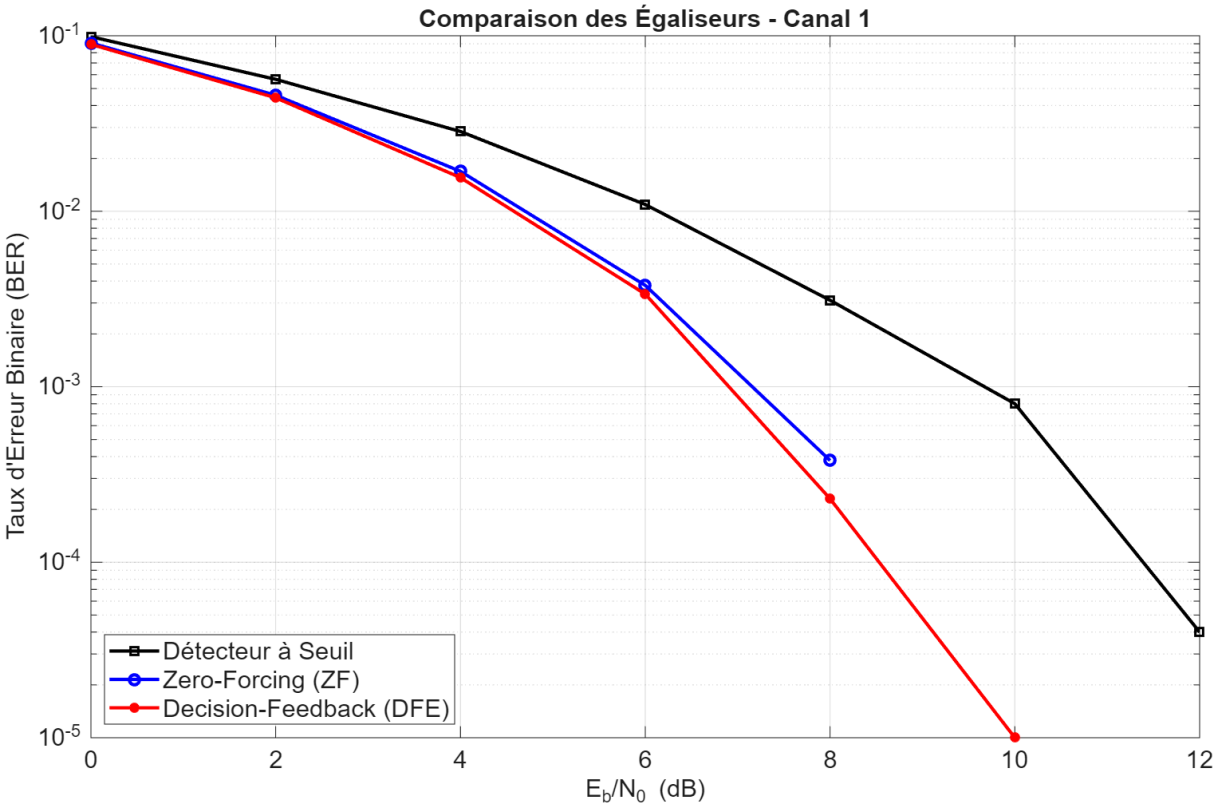

Figure 10: Comparison of the three equalizers for BPSK on channel 1

Interpretation (Channel 1): As predicted, the weak echoes allow ZF and DFE to achieve excellent and nearly identical performance. The threshold detector holds at low SNR but shows a nascent error floor around 8 dB.

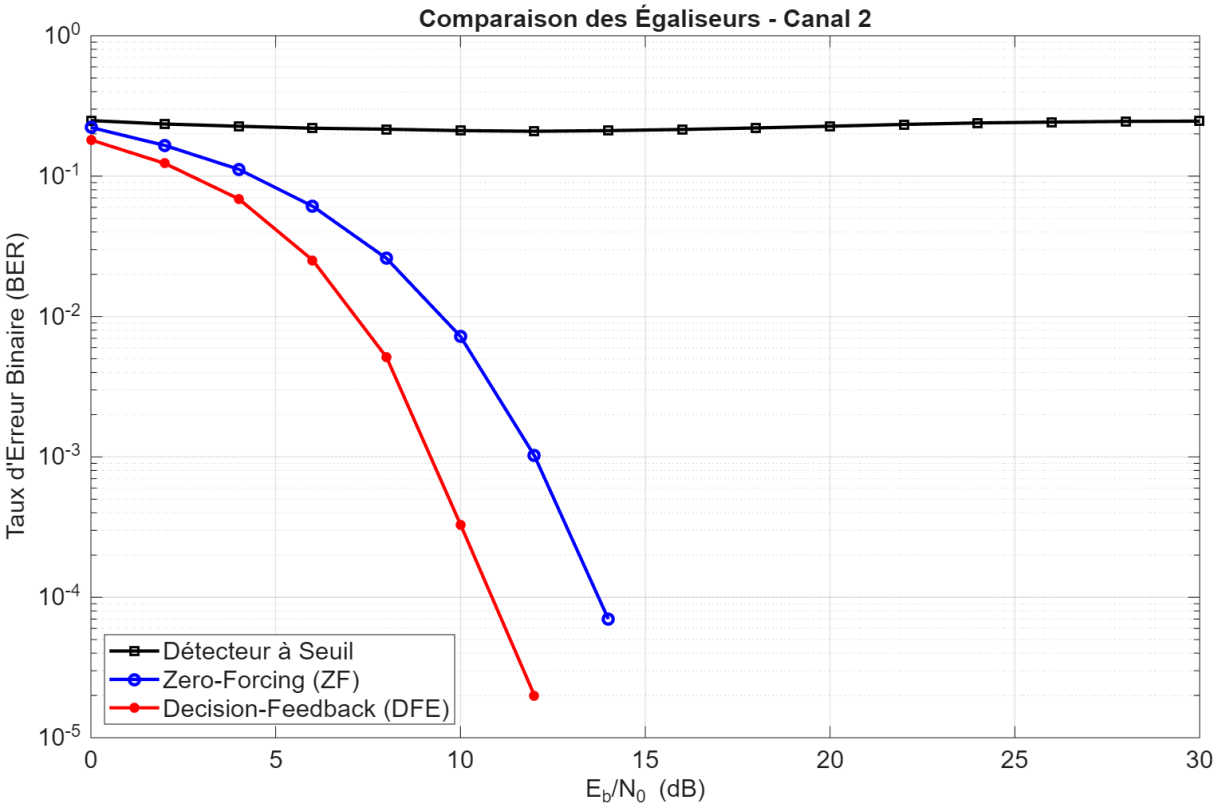

Figure 11: Comparison of the three equalizers for BPSK on channel 2

Interpretation (Channel 2): The significant ISI completely closes the eye of the signal, rendering the threshold detector inoperative (BER constant around 2.10^{-1}). The ZF manages to decode the signal but suffers a penalty of approximately 2.5 dB compared to the DFE to reach a BER of 10^{-4}. The DFE demonstrates the advantage of its feedback loop here.

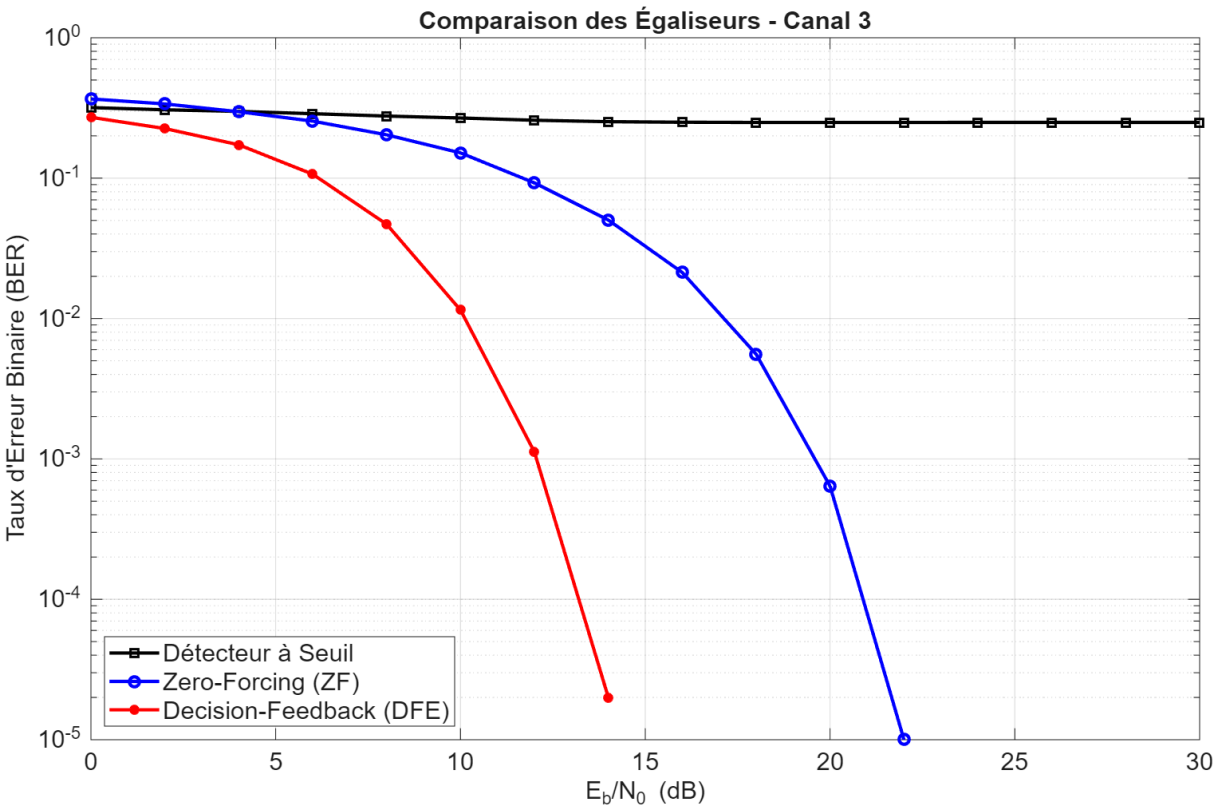

Figure 12: Comparison of the three equalizers for BPSK on channel 3

Interpretation (Channel 3): On this extremely difficult channel, the ZF collapses due to excessive noise amplification (it requires 22 dB for a BER of 10^{-5}). The DFE demonstrates its great algorithmic superiority by reaching this same BER from 14 dB, representing a massive performance gain of 8 dB over the ZF.

3.5 Simulation Results and Interpretations (8-QAM Modulation)

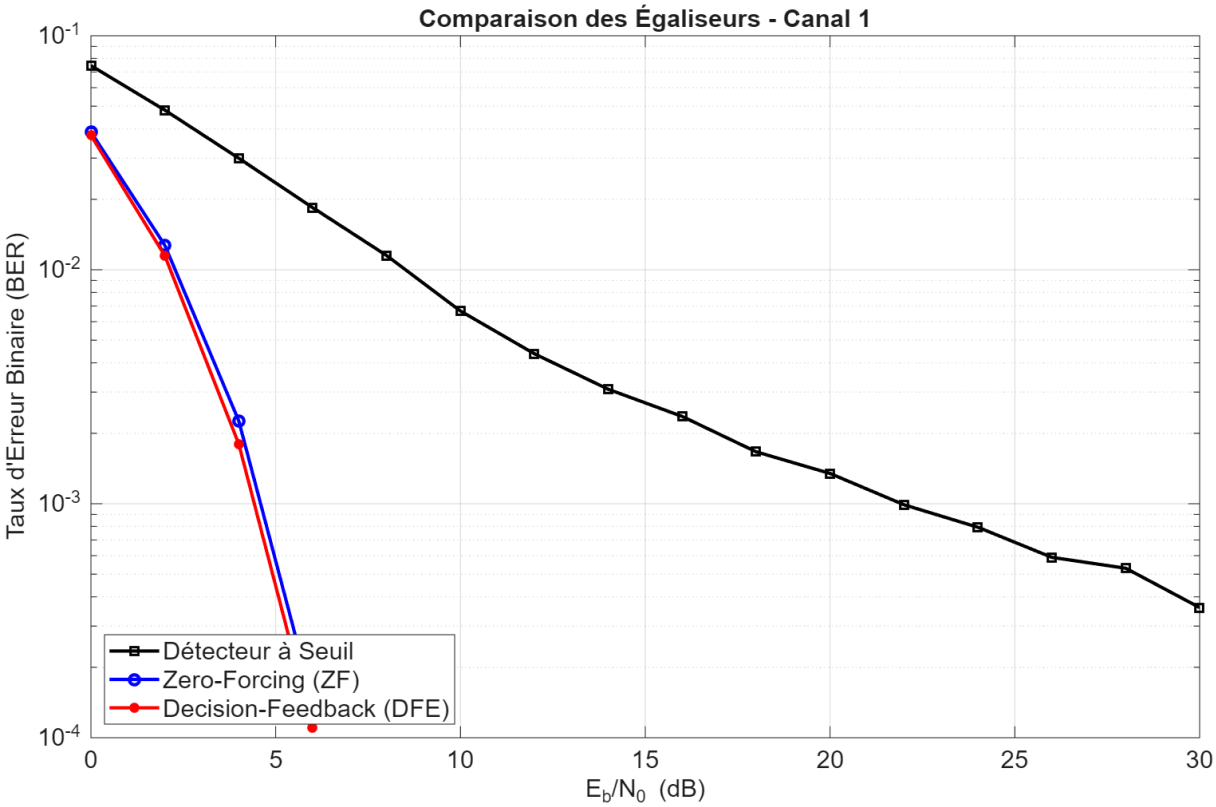

Figure 13: Comparison of the three equalizers for 8-QAM on channel 1

Interpretation (Channel 1): The 8-QAM modulation has a denser constellation than BPSK, with reduced inter-symbol distances. Consequently, it is much more sensitive to ISI. Even though the echoes of channel 1 are weak, they are sufficient to strongly degrade the performance of the threshold detector compared to BPSK. The ZF and DFE, on the other hand, correct this slight ISI without significantly amplifying the noise, displaying nearly superimposed curves.

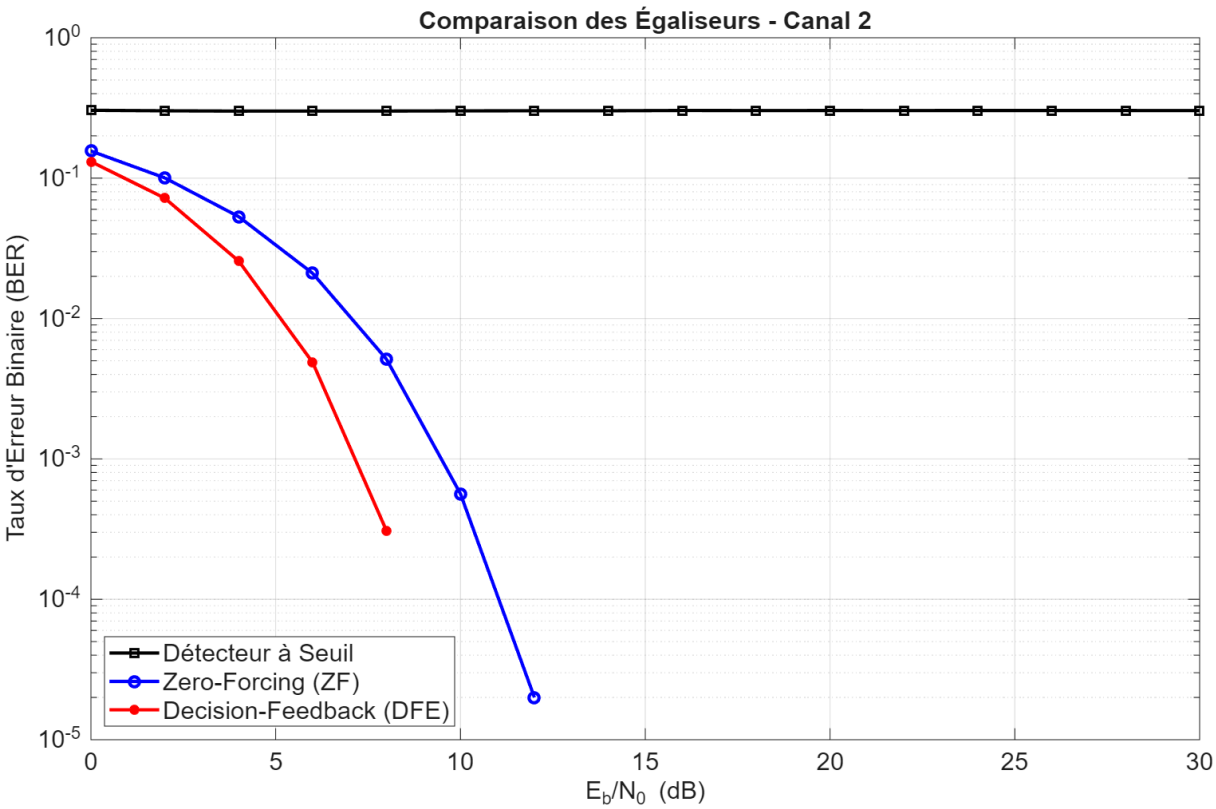

Figure 14: Comparison of the three equalizers for 8-QAM on channel 2

Interpretation (Channel 2): The significant ISI completely closes the eye of the signal, rendering the threshold detector completely inoperative (BER constant around 3.10^{-1}). The high sensitivity of 8-QAM highlights the noise amplification suffered by the ZF: it requires approximately 11.5 dB to achieve a BER of 10^{-4}, while the DFE reaches it from 9 dB. The DFE's decision feedback approach demonstrates its robustness here when faced with a complex constellation.

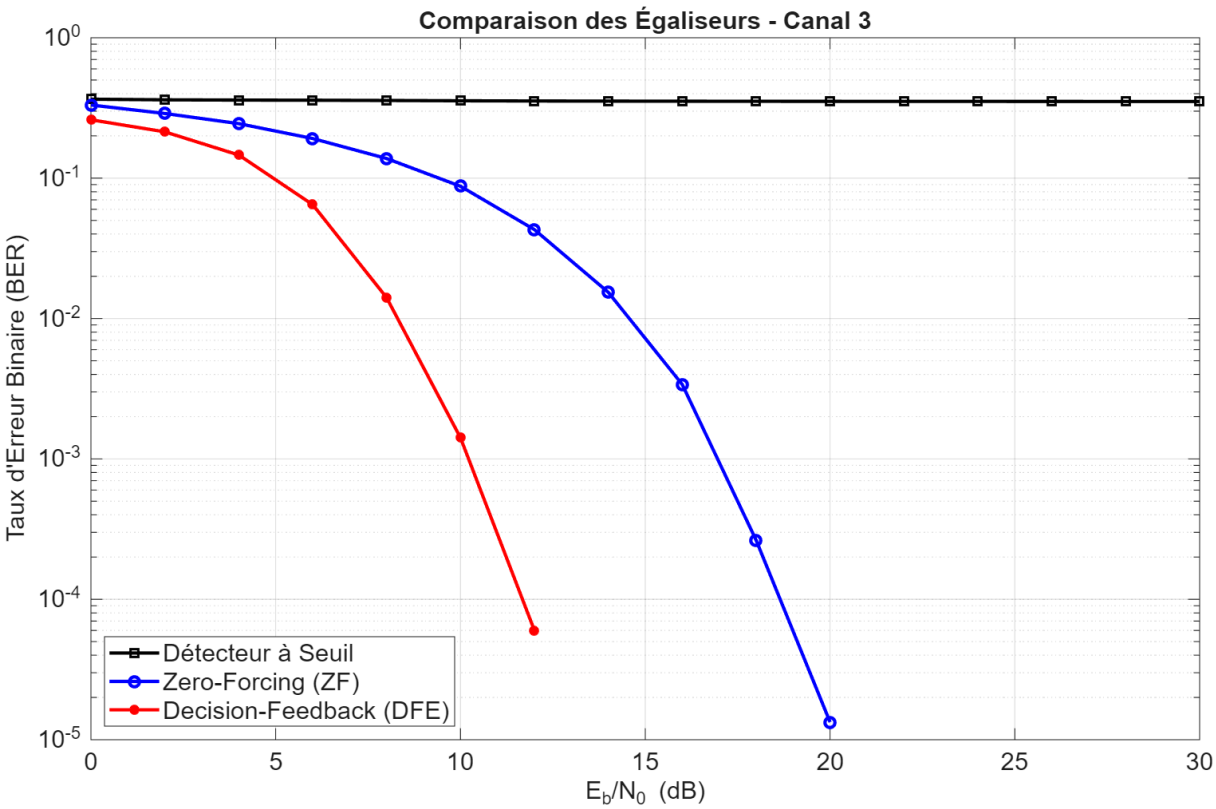

Figure 15: Comparison of the three equalizers for 8-QAM on channel 3

Interpretation (Channel 3): On this channel with very deep fading, the channel matrix is poorly conditioned. The ZF's weaknesses are further exacerbated by the density of the 8-QAM constellation: channel inversion drastically amplifies noise, pushing the BER of 10^{-4} to nearly 19 dB. The DFE elegantly bypasses this mathematical inversion through QR decomposition and its feedback loop, reaching this same quality of service from 11.5 dB. This significant gain of 7.5 dB demonstrates the absolute necessity of a DFE equalizer for higher-order modulations in highly selective environments.

3.6 Simulation Results and Interpretations (16-QAM Modulation)

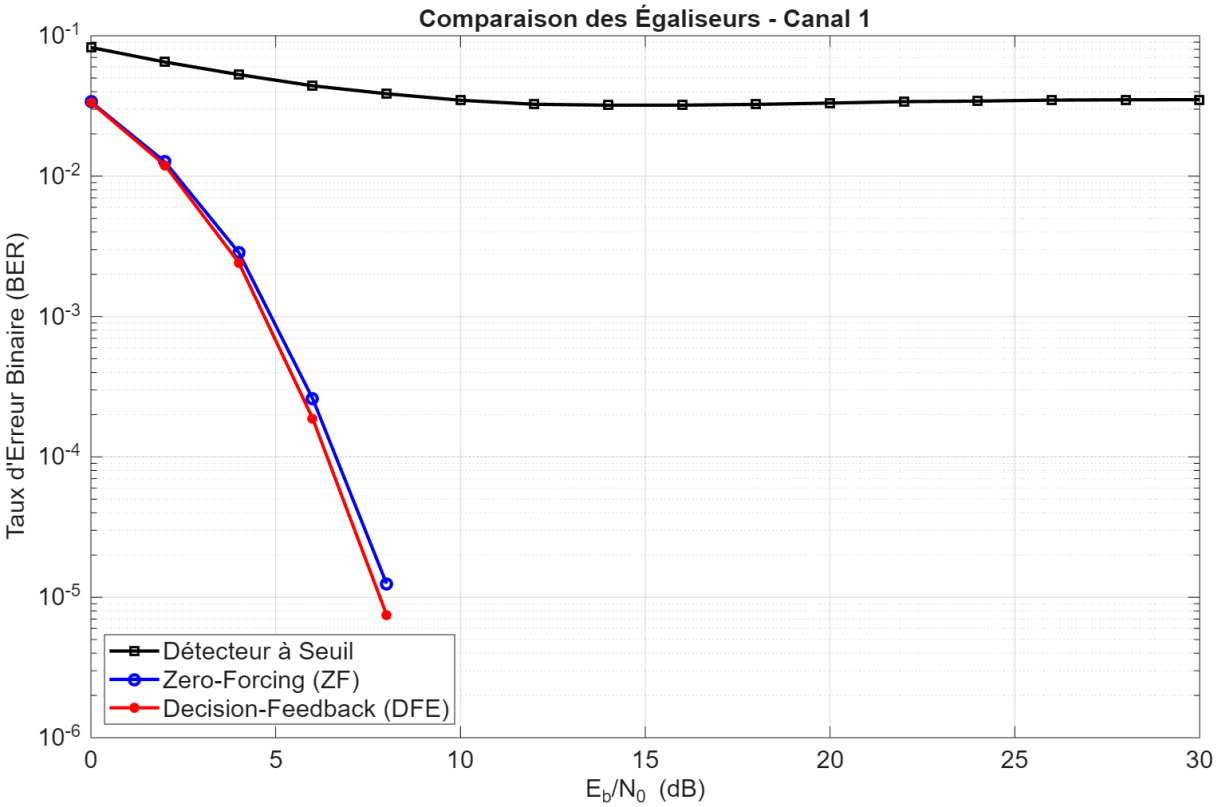

Figure 16: Comparison of the three equalizers for 16-QAM on channel 1

Interpretation (Channel 1): Increasing the modulation order to 16 (16-QAM) considerably reduces the Euclidean distance between constellation symbols. The major observation here is the total collapse of the threshold detector (BER stuck around $4.10^{\{-2\}}$). Unlike BPSK, the accumulation of weak echoes (0.1) is now sufficient to push symbols across decision boundaries. However, since the matrix H remains well-conditioned, the ZF and DFE easily invert the channel without amplifying noise, offering optimal and superimposed performance.

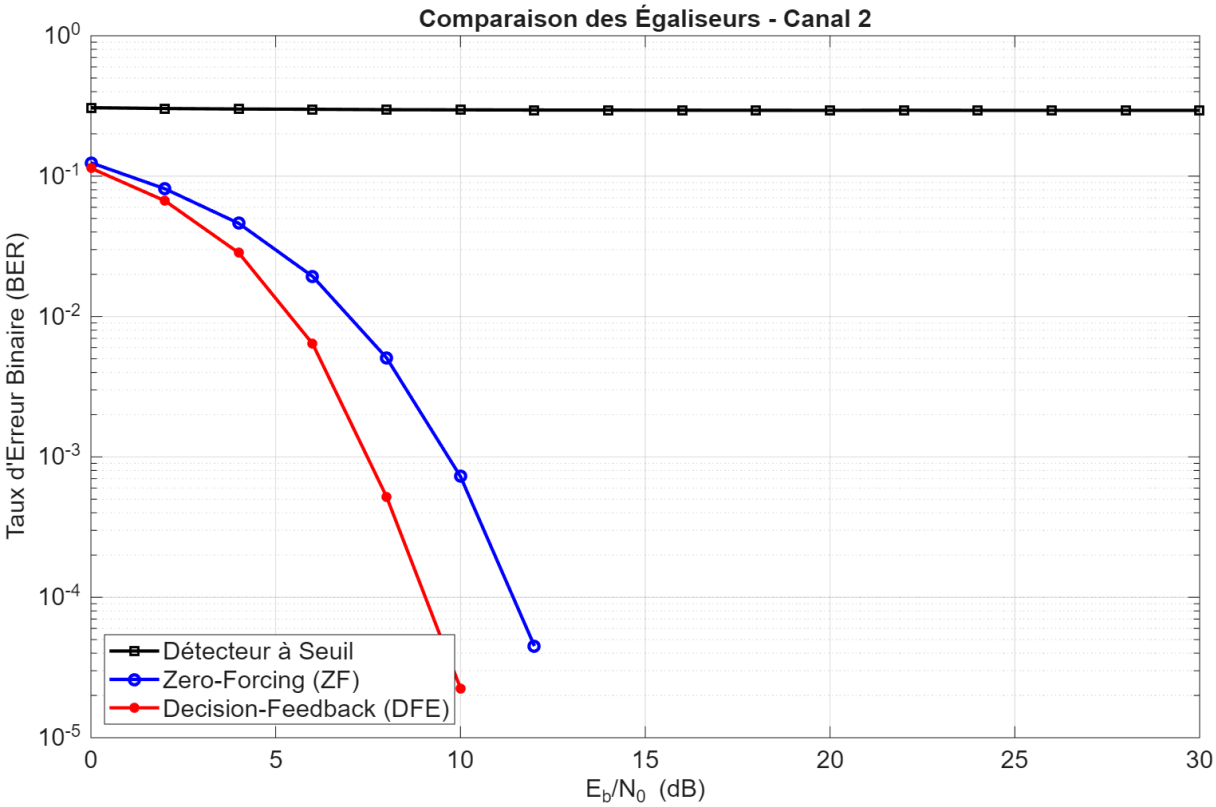

Figure 17: Comparison of the three equalizers for 16-QAM on channel 2

Interpretation (Channel 2): With very strong Inter-Symbol Interference, the threshold detector logically remains inoperative (BER $\approx 3.10^{-1}$). With a very dense constellation, the noise amplification caused by the Zero-Forcing (ZF) equalizer becomes even more penalizing: it requires approximately 11.5 dB to achieve a BER of 10^{-4}. The DFE handles ISI elimination much better thanks to its decision feedback loop, reaching this same error rate with an advance of approximately 2.5 dB.

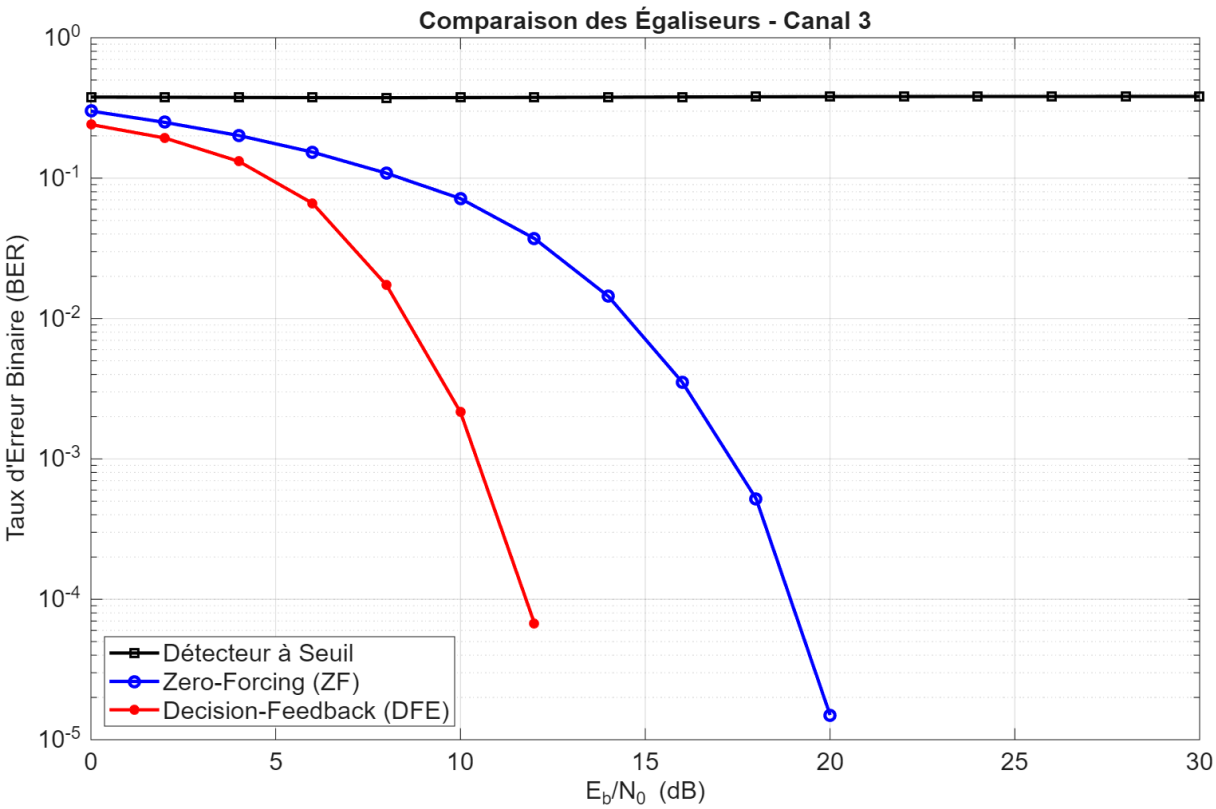

Figure 18: Comparison of the three equalizers for 16-QAM on channel 3

Interpretation (Channel 3): On this deep fading channel (very strong echoes), the Toeplitz matrix H is very poorly conditioned. The matrix inversion of the ZF causes catastrophic noise amplification, which is fatal for a modulation as sensitive as 16-QAM: the curve falls very slowly, requiring nearly 19 dB to reach a BER of 10^{-4}. The DFE confirms its absolute superiority here by bypassing direct channel inversion, reaching this same performance of 10^{-4} from 11.5 dB, representing a gain of 7.5 dB over the ZF.

3.7 Overall Performance Summary

The analysis highlights three key lessons for communications over frequency-selective channels.

First, as the modulation order increases (from BPSK to 16-QAM), the system becomes more sensitive to ISI. The reduction in inter-symbol distance renders simple detectors ineffective from the presence of weak echoes, especially with dense constellations.

Second, the Zero-Forcing linear equalizer shows its limits: performing well on weakly dispersive channels, it becomes penalizing on severe channels. Inverting a poorly conditioned channel matrix leads to strong noise amplification, which greatly degrades performance.

Finally, the Decision Feedback Equalizer (DFE) emerges as the most robust solution. Thanks to its feedback loop, it suppresses ISI without amplifying noise, offering significant gains (up to 8 dB on the most severe channels). It thus becomes indispensable to guarantee a BER $\leq 10^{-4}$ with high-order modulations in hostile environments.

3.8 Extra credits

We now seek to combine the different parts: what are the performances of the different detectors in the case of coding.

We return to a coded (or uncoded) BPSK modulation on the three previous channels.

3.8.1 For the first channel (weakly noisy)

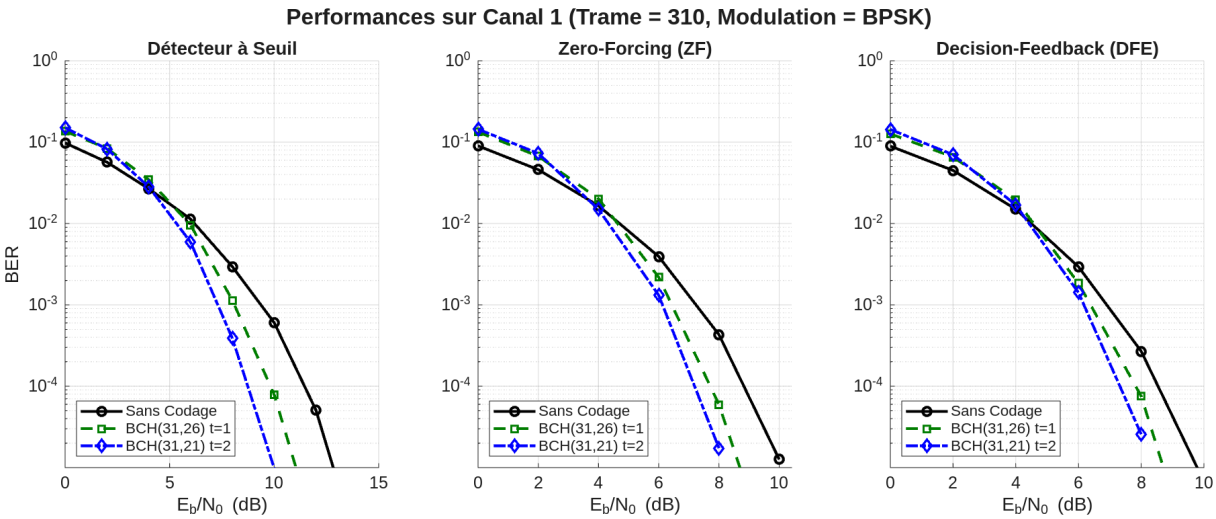

Figure 19: Comparison of detectors on channel 1 (weakly noisy)

- **Threshold detector**: The Inter-Symbol Interference (ISI) is sufficiently weak for the eye to remain partially open, but a nascent error floor is observed at high $\frac{E_b}{N_0}$. The addition of BCH codes effectively cleans up these few residual errors, restoring a pronounced downward slope to the curve.
- **ZF and DFE Equalizers**: On this mild channel, the matrix H is very well conditioned. ISI is cancelled without notable amplification of thermal noise. Consequently, the ZF and DFE equalizer curves are nearly superimposed and behave as on a pure AWGN channel. Integrating BCH codes provides the expected theoretical coding gain here (approximately 2 to 3 dB gain for t=2 at a BER of 10^{-4}).

3.8.2 For the second channel (noisy)

The moderate echoes (up to 0.8) severely spread symbol energy. The interactions between the equalizer and the decoder become critical here.

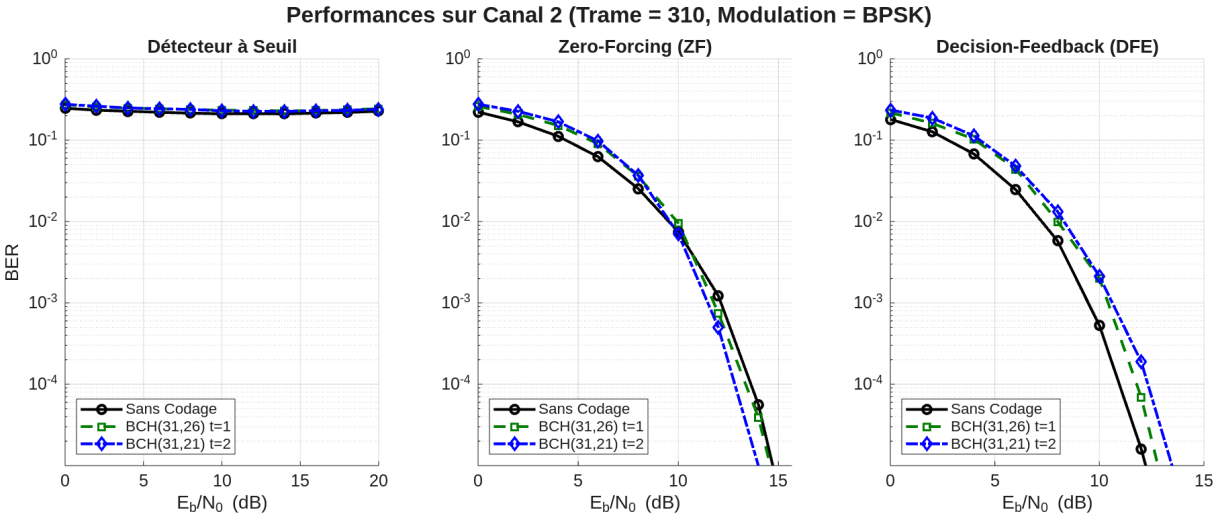

Figure 20: Comparison of detectors on channel 2 (noisy)

- **Threshold Detector**: The raw error rate stagnates around 2.10^{-1}. At this level, the channel generates far more errors (20%) than the correction capacity of BCH codes (which can only correct 3 to 6% of bits). The decoders are completely saturated: adding coding brings no improvement, the curve remains a flat floor.
- **Zero-Forcing (ZF) Equalizer**: Without coding, the ZF suffers from noise amplification due to channel matrix inversion. However, this amplification generates purely white (Gaussian) noise. The resulting errors are therefore random and decorrelated — which is exactly the use case for a block code. An interesting compensation is observed here: the coding gain provided by BCH2 is significant (nearly 4 dB of gain) and largely compensates for the structural weakness of the ZF.
- **DFE Equalizer**: The uncoded DFE outperforms the uncoded ZF. However, it is noted that the gain provided by BCH is proportionally smaller on the DFE than on the ZF. This phenomenon is explained by the nature of the DFE's feedback loop: when a decision error occurs, it is fed back into the filter, causing bursts of consecutive errors (burst errors). BCH codes, designed for isolated errors, struggle to correct these grouped errors within a single 31-bit codeword.

3.8.3 For channel 3 (very noisy)

This channel is the most hostile, with all echoes having strong amplitude (0.8), making the matrix H extremely poorly conditioned.

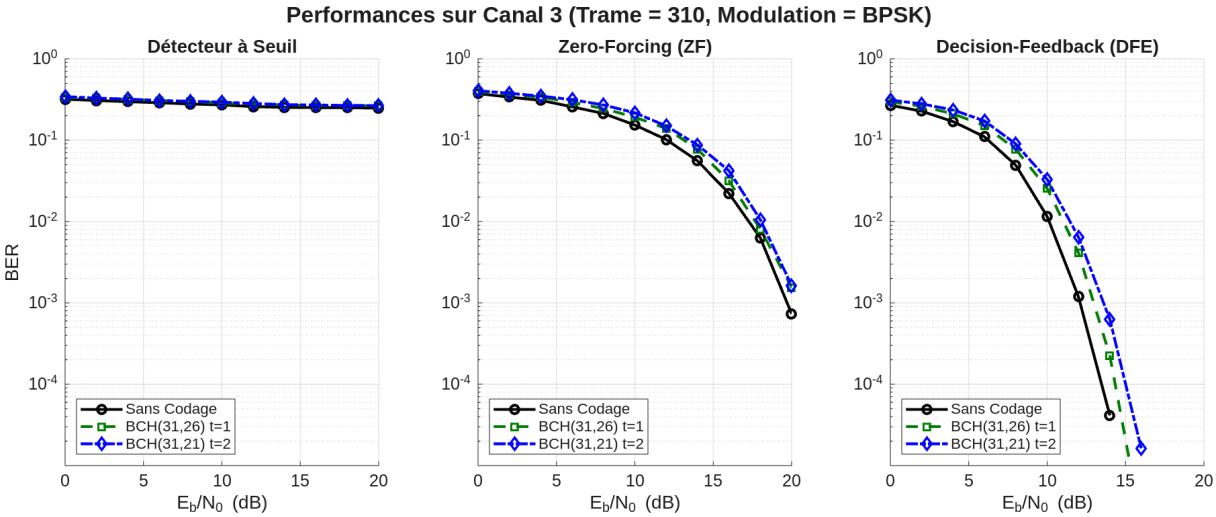

Figure 21: Comparison of detectors on channel 3 (very noisy)

- **Threshold Detector**: Unsurprisingly, it remains completely inoperative (BER stagnating at 3.10^{-1}) and BCH codes are useless here.
- **Zero-Forcing (ZF) Equalizer**: The noise amplification phenomenon becomes very significant. The uncoded version requires a very good $\frac{E_b}{N_0}$ to converge. In this context, channel coding becomes necessary. The BCH t=2 code "saves" the ZF algorithm, bringing the curve to more acceptable proportions through the correction of decorrelated errors, although overall performance remains moderate.
- **DFE Equalizer**: It confirms its algorithmic superiority here. Bypassing the poor matrix conditioning through its QR factorization and past decisions, it offers the best raw performance. Even if the burst error phenomenon limits the full theoretical potential of the BCH decoder, the combination DFE + BCH(31,21) t=2 constitutes by far the most robust architecture of all simulations. It is the only one capable of guaranteeing extremely reliable communication ($BER < 10^{-5}$) at reasonable power (around 13dB) on such a severe propagation channel.

3.8.4 Throughput measurement for different configurations

To further characterize the performance of different coding and detection configurations across channels, we now introduce the concept of "throughput".

- The throughput is defined as

$$T = FSR \frac{k}{\frac{n}{\log 2(C)} T_s}$$

with:

- FSR (Frame Success Rate): the probability that no information bit in the frame is erroneous
- k: the number of information bits per frame
- n: the number of symbols per frame
- Ts: the symbol duration
- C: the size of the modulation constellation

A frame is erroneous if at least one of the k information bits is in error, i.e. $FSR = (1 - BER)^{\mathrm{k}}$

From the above:

- without coding $BER = Q\left(\sqrt{\frac{2Es}{N_0}}\right)$
- with BCH coding, $BER = \frac{1}{k}\sum_{j=t+1}^{n}\binom{n}{j}p^{\mathrm{j}}\,(1-p)^{\mathrm{n\text{-}j}}$

Finally, $T = k\frac{\log 2(C)}{nT_s}(1 - BER)^{\mathrm{k}}$ with the BER adapted to the coding choice.

After simulation, we obtain the following curves:

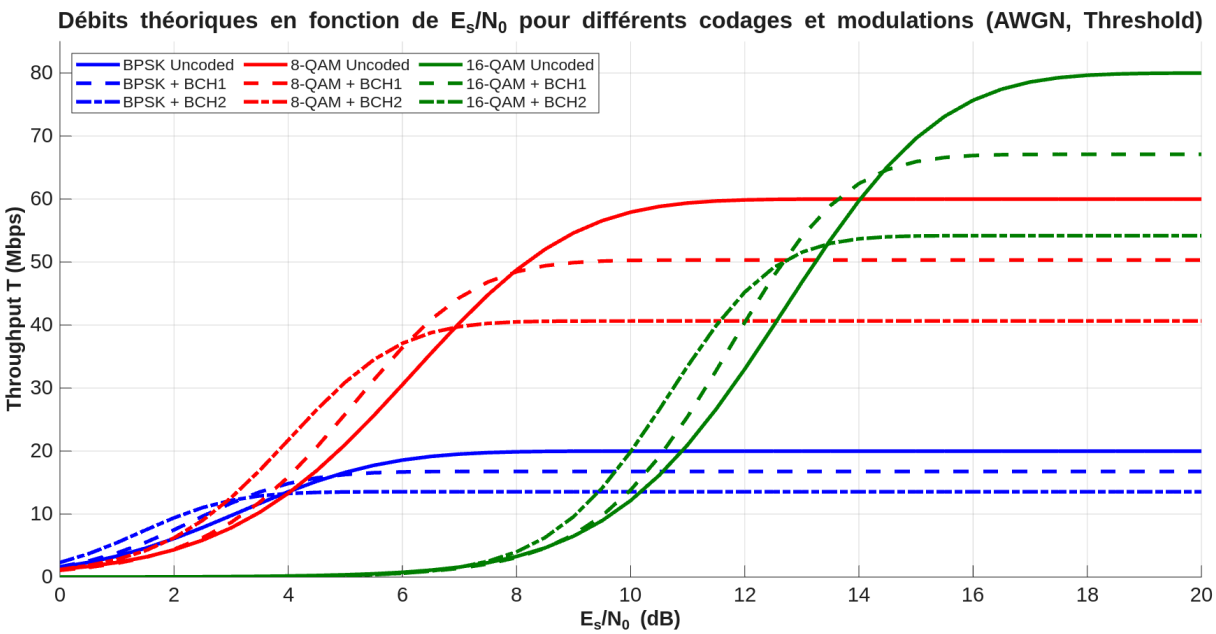

Figure 22: Throughput comparison for a given SNR as a function of the transmitter/ receiver chain configuration

Dense as it is, this graph allows, once the propagation channel is characterized, to choose the right configuration according to the throughput or SNR specified by the requirements.

It can nevertheless be noted that, at low SNR, throughput collapses completely, the channel being extremely noisy. At high SNR, on the other hand, it is the coding and modulation choices that limit communication performance.

This graph proves the importance of “Adaptive Modulation and Coding” in 4G and 5G, based on signal quality.